PHYSICS

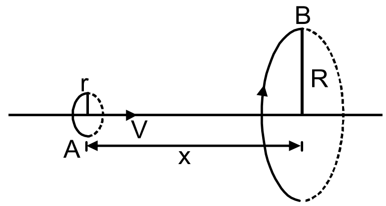
TARGET : JEE- Advanced

CAPS-15

EMI & AC

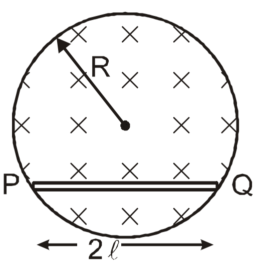
SCQ (Single Correct Type) :

1. Loop A of radius r(r << R) moves towards loop B with a constant velocity V in such a way that their planes are always parallel. What is the distance between the two loops (x) when the induced emf in loop A is maximum



(A) R (B) $\frac{R}{\sqrt{2}}$ (C) $\frac{R}{2}$ (D) $R\left(1-\frac{1}{\sqrt{2}}\right)$

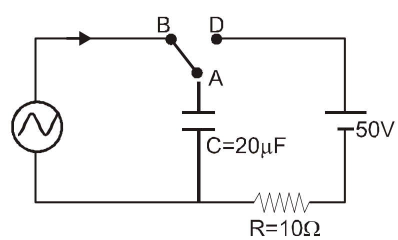
2. A uniform magnetic field, $B = B_0 t$ (where B_0 is a positive constant), fills a cylindrical volume of radius R, then the potential difference in the conducting rod PQ due to electrostatic field is :



(A) $B_0\ell\sqrt{R^2+\ell^2}$ (B) $B_0\ell\sqrt{R^2-\frac{\ell^2}{4}}$ (C) $B_0\ell\sqrt{R^2-\ell^2}$ (D) $B_0R\sqrt{R^2-\ell^2}$

MCQ (One or more than one correct) :

3. At time t = 0, terminal A in the circuit shown in the figure is connected to B by a key and alternating current $I(t) = I_0\cos(\omega t,)$, with I_0 = 1A and ω = 500 rad s^{-1} starts flowing in it with the initial direction shown in the figure. At t = $\frac{7\pi}{6\omega}$, the key is switched from B to D. Now onwards only A and D are connected. A total charge Q flows from the battery to charge the capacitor fully. If C = 20μ, R = 10Ω and the battery is ideal with emf of 50V, identify the correct statement (s)



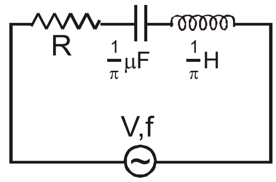
(A) Magnitude of the maximum charge on the capacitor before $t = \frac{7\pi}{6\omega}$ is 1×10^{-3} C.

(B) The current in the left part of the circuit just before $t = \frac{7\pi}{6\omega}$ is clockwise

(C) Immediately after A is connected to D. the current in R is 10A.

(D) $Q = 2 \times 10^{-3}$ C.

4. In the AC circuit shown below, the supply voltage has constant rms value V but variable frequency f. At resonance, the circuit :



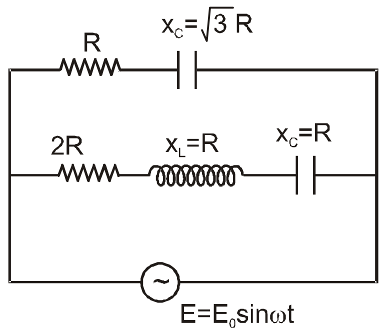
(A) has a current I given by $I = \frac{V}{R}$

(B) has a resonance frequency 500 Hz

(C) has a voltage across the capacitor which is 180^0 out of phase with that across the inductor

(D) has a current given by $I = \frac{V}{\sqrt{R^2 + \left(\frac{1}{\pi} + \frac{1}{\pi}\right)^2}}$

5. Choose the correct options for the given arrangement (Symbols have their usual meaning)



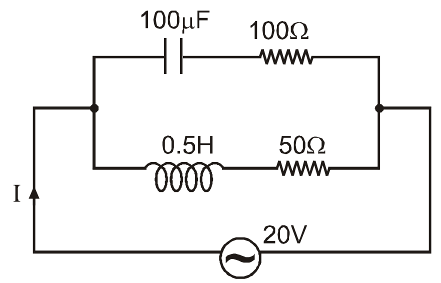
(A) Current from the AC source is $\frac{\pi}{6}$ phase ahead than voltage.

(B) Impendence of the circuit is $\frac{2R}{\sqrt{3}}$

(C) Current from resistor R as function of time is $\frac{E_0}{2R}\sin\left(\omega t + \frac{\pi}{3}\right)$

(D) Current from resistor 2R as function of time is $\frac{E_0}{2R}\sin\omega t$

6. In the given circuit, the AC source has ω = 100 rad/s. considering the inductor and capacitor to be ideal, the correct choice (s) is(are)



(A) The current through the circuit, I is approximately 0.3 A

(B) The current through the circuit, I is $0.3\sqrt{2}$ A.

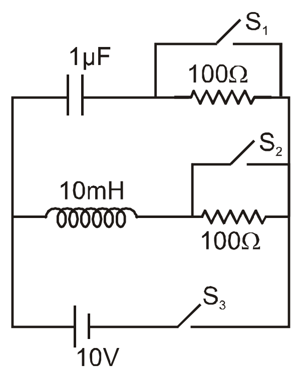
(C) The voltage across 100Ω resistor = $10\sqrt{2}$ V

(D) The voltage across 50Ω resistor = 10V

Comprehension Type Question:

Comprehension # 1

Switches S_1 and S_2 are open while S_3 remains closed for long time such that capacitor becomes fully charged and current in inductor coil becomes maximum. Now switches S_1 and S_2 are closed and at the same time S_3 is opened at t = 0. Assume that battery and inductor coil are ideal.



7. Charge (in μC) on capacitor as a function of time is :

(A) $q = 10\sqrt{2}\ \sin\left(10^4 t(\text{sec.}^{-1}) + \frac{\pi}{4}\right)$

(B) $q = 10 \sin\left(10^4 t(\text{sec.}^{-1}) + \frac{\pi}{2}\right)$

(C) $q = 10 \sin\left(10^4 t(\text{sec.}^{-1}) + \frac{\pi}{4}\right)$

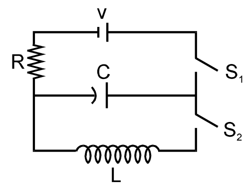
(D) $q = 10\sqrt{2} \sin\left(10^2 t(\text{sec.}^{-1}) + \frac{\pi}{4}\right)$

8. Maximum current in the inductor coil at any time t > 0 is :

(A) $\frac{1}{10}$ A (B) $\frac{1}{5\sqrt{2}}$ A (C) $\frac{1}{5}$ A (D) 1 A

Comprehension # 2

The capacitor of capacitance C can be charged (with the help of a resistance R) by a voltage source V, by closing switch S_1 while keeping switch S_2 open. The capacitor can be connected in series with an inductor 'L' by closing switch S_2 and opening S_1.



9. Initially, the capacitor was uncharged. Now, switch S_1 is closed and S_2 is kept open. If time constant of this circuit is τ, then

(A) after time interval τ, charge on the capacitor is CV/2

(B) after time interval 2τ, charge on the capacitor is $CV(1 - e^{-2})$

(C) the work done by the voltage source will be half of the heat dissipated when the capacitor is fully charged

(D) after time interval 2τ, charge on the capacitor is $CV(1 - e^{-1})$

10. After the capacitor gets fully charged, S_1 is opened and S_2 is closed so that the inductor is connected in series with the capacitor. Then,

(A) at t = 0, energy stored in the circuit is purely in the form of magnetic energy

(B) at any time t > 0, current in the circuit is in the same direction

(C) at t > 0, there is no exchange of energy between the inductor and capacitor

(D) at any time t > 0, instantaneous current in the circuit will have maximum value $V\sqrt{\frac{C}{L}}$, where C is the capacitance and L is the inductance.

11. If the total charge stored in the LC circuit is Q_0, then for $t \geq 0$

(A) the charge on the capacitor is $Q = Q_0 \cos\left(\frac{\pi}{2} + \frac{t}{\sqrt{LC}}\right)$

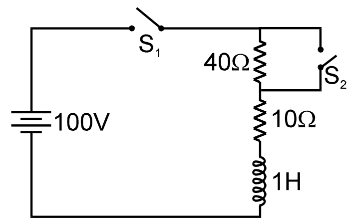
(B) the charge on the capacitor is $Q = Q_0 \cos\left(\frac{\pi}{2} - \frac{t}{\sqrt{LC}}\right)$

(C) the charge on the capacitor is $Q = -LC\,\frac{d^2Q}{dt^2}$

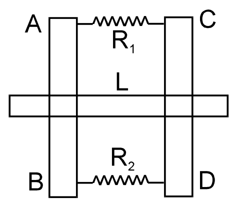
(D) the charge on the capacitor is $Q = -\frac{1}{\sqrt{LC}}\,\frac{d^2Q}{dt^2}$

Numerical based Questions :

12. In the circuit diagram shown in the figure the switches S_1 and S_2 are closed at time t = 0. After time t = (0.1) ℓn 2 sec, switch S_2 is opened. The current in the circuit at time, t = (0.2) l n 2 sec is equal to $\frac{x}{32}$ amp. Findout value of x.

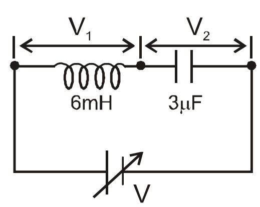
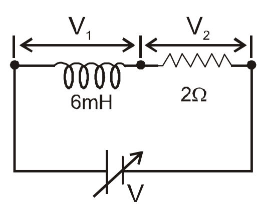
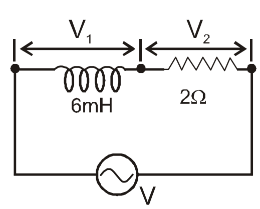


13. Two parallel vertical metallic rails AB and CD are separated by 1 m. They are connected at the two ends by resistance R_1 and R_2 as shown in the figure. A horizontal metallic bar L of mass 0.2 kg slides without friction, vertically down the rails under the action of gravity. There is a uniform horizontal magnetic field of 0.6T perpendicular to the plane of the rails. It is observed that when the terminal velocity is attained, the power dissipated in R_1 and R_2 are 0.76 W and 1.2 W respectively. If the terminal velocity of bar L is x m/s and R_1 is y Ω and R_2 is ZΩ then find the value of $x + 76y + 10z$. ($g = 9.8$ m/s^2)

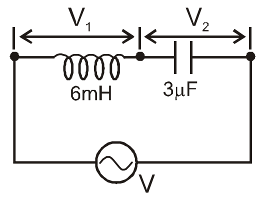


Matrix Match Type :

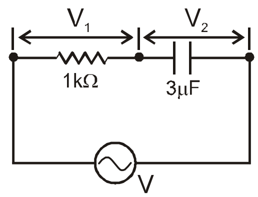
14. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in different ways as shown in **Column II.** When a current I (steady state for DC or rms for AC) flows through the circuit, the corresponding voltage V_1 and V_2. (indicated in circuits) are related as shown in **Column I.** Match the two column.

Column I	**Column II**
(A) $I \neq 0, V_1$ is proportional to I	(p) 
(B) $I \neq 0, V_2 > V_1$	(q) 
(C) $V_1 = 0, V_2 = V$	(r) 

(D) $I \neq 0, V_2$ is proportional to I (s)



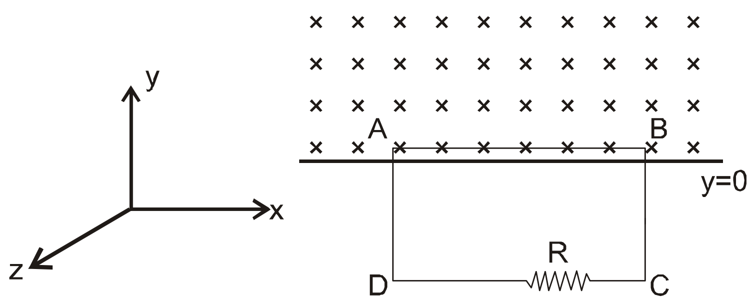
(t)



Subjective Type Questions :

15. A metal disc of radius R, resistivity ρ, thickness y is placed in a vertical magnetic field of induction $B = B_0 \sin \omega t$ where $\omega = 2\pi f$. Find the total eddy current loss in the metal disc.

16. A square loop ABCD of side ℓ is moving in xy plane with velocity $\vec{v} = \beta t\ \hat{j}$. There exists a non-uniform magnetic field $\vec{B} = -B_0(1 + \alpha y^2)\ \hat{k}$ (y > 0), where B_0 and α are positive constants. Initially, the upper wire of the loop is at y = 0. Find the induced voltage across the resistance R as a function of time. Neglect the magnetic force due to induced current.



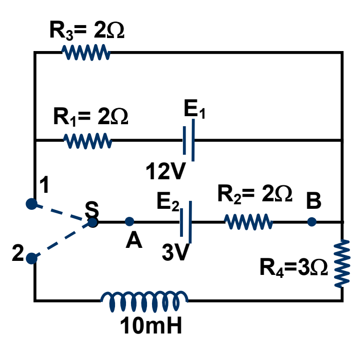
17. A circuit containing a two position switch S is shown below.

(a) The switch S is in position 1. Find the potential difference $V_A - V_B$ and rate of production of heat in R_1.

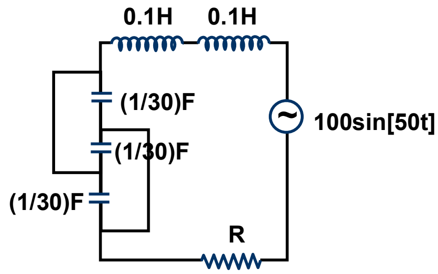
(b) If now the switch S is put in position 2 at t = 0, find

(i) steady current in R_4.

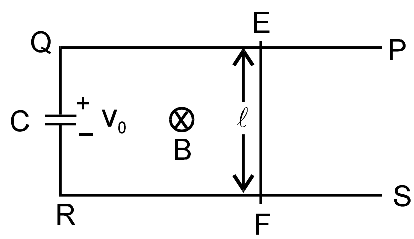
(ii) The time when the current in R_4 is half the steady value. Also, calculate the energy stored in the inductor L at that time.



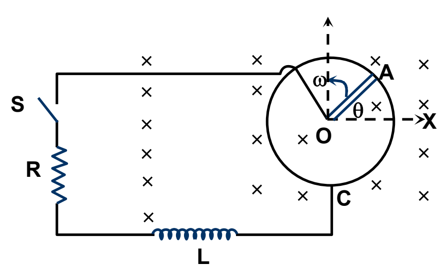
18. Find the value of the resistance R so that the power factor of the given circuit is $\frac{1}{\sqrt{2}}$. Also find the peak current in this case.



19. In the figure shown 'PQRS' is a fixed resistanceless conducting frame in a uniform and constant magnetic field of strength B. A rod 'EF' of mass 'm', length 'ℓ' and resistance R can smoothly move on this frame. A capacitor charged to a potential difference 'V_0' initially is connected as shown in the figure. Find the velocity of the rod as function of time 't' if it is released at t = 0 from rest.



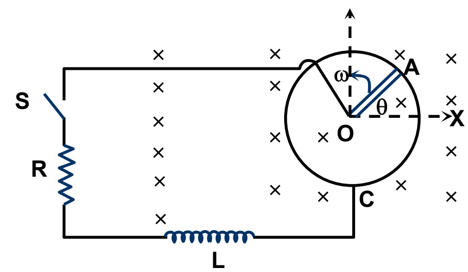
20. A metal rod OA of mass m and length r is kept rotating with a constant angular speed ω in a vertical plane about a horizontal axis at the end O. The free end A is arranged to slide without friction along a fixed conducting circular ring in the same plane as that of rotation. A uniform and constant magnetic induction $\vec{B}$ is applied perpendicular and into the plane of rotation as shown in figure. An inductor L and an external resistance R are connected through a switch S between the point O and a point C on the ring to form an electrical circuit. Neglect the resistance of the ring and the rod. Initially, the switch is open. What is the induced emf across the terminals of the switch?



21. In the previous question the switch S is closed at time t = 0.

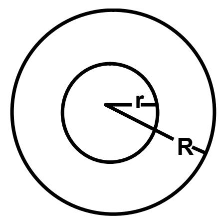
(i) Obtain an expression for the current as a function of time.

(ii) In the steady state, obtain the time dependence of the torque required to maintain the constant angular speed. Given that the rod OA was along the positive X–axis at t = 0.

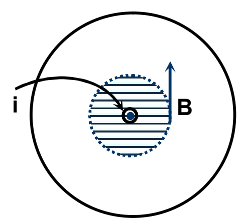


22. An infinitesimal bar magnet of dipole moment M is pointing and moving with speed v in the x-direction. A closed circular conducting loop of radius a and negligible self inductance lies in the Y-Z plane with its center at x = 0 and its axis coinciding with x-axis. Find the force opposing the motion of the magnet, if the resistance of the loop is R. Assuming $x \gg R$.

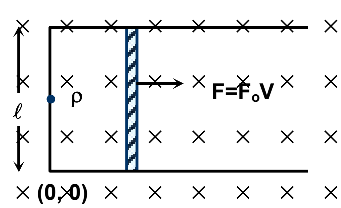
23. Find the magnetic energy stored in the system of two concentric loops of radii r and R (R>>r). Find the mutual inductance of the system. If the loops have currents of same magnitude i_0, find the magnetic energy stored. Ignore the effect of self inductance



24. A long solid cylindrical straight conductor carries a current i_0. Find the magnetic energy stored per meter inside it.



25. Two infinite parallel wires, having the cross sectional area a and resistivity k are connected at a junction point P (as shown in the figure). A slide wire of negligible resistance and having mass 'm' and length 'ℓ' can slide between the parallel wires, without any frictional resistance.



If the system of wires is introduced to a magnetic field of intensity B (into the plane of paper) and the slide wire is pulled with a force which varies with the velocity of the slide wire as $F = F_0V$, then find the velocity of the slide wire as a function of the distance travelled. (The slide wire is initially at origin and has a velocity v_0)

ANSWER KEY (CAPS-15)

1.	(C)	**2.**	(C)	**3.**	(CD)	**4.**	(ABC)		
5.	(ABCD)	**6.**	(AC)	**7.**	(A)	**8.**	(B)	**9.**	(B)
10.	(D)	**11.**	(C)	**12.**	67	**13.**	40		

14. (A) – r,s,t ; (B) – q,r,s,t ; (C) – p,q ; (D) – q,r,s,t

15. $P = \dfrac{\pi^3 B_0^2 f^2 R^4 y}{4\rho}$

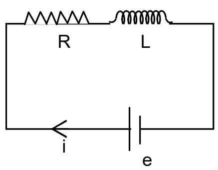
16. $\varepsilon = -B_0 I\beta\left(t + \dfrac{\alpha \times \beta^2 t^5}{4}\right)$

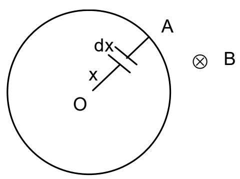
17. $E = \dfrac{1}{2}LI^2 = \dfrac{1}{2} \times 10 \times 10^{-3} \times (0.3)^2 J = 4.5 \times 10^{-4} J$

18. $\therefore$ Peak current $= \dfrac{100}{9\cdot 8 \times \sqrt{2}} = 7\cdot 22$ Amperes.

19. $v = \dfrac{B\ell C \;\; V_0}{m + B^2\ell^2 C}\left(1 - e^{-\left(\frac{B^2\ell^2}{mR} + \frac{1}{RC}\right)t}\right)$

20. $de = B\,(x\omega)\,dx \qquad \Rightarrow e = \dfrac{B\omega r^2}{2}$





21. $\tau = \dfrac{mgr\cos\omega t}{2} + \dfrac{1}{4R}B^2\omega r^4$

22. $F = \dfrac{9\mu_0^2 M^2 a^4}{4x^8 R}v$

23. $U = \dfrac{\pi\mu_0 r^2 i_0^2}{2R}$ $(\because i_1 = i_2 = i_0)$

24. $U = \dfrac{\mu_0 i_0^2 \ell}{4\pi} \times \dfrac{1}{4} = \dfrac{\mu_0 i_0^2}{16\pi}$ $(\because \ell = 1)$

25. $\dfrac{F_0}{m}x - \dfrac{B^2\ell^2 a}{2km}\left\{\ln\left(\dfrac{2x+\ell}{\ell}\right)\right\} + v_o = V$